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Intrusion and Magnetization of the Giles Complex, Central Australia

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Summary

From a study of five hundred and forty specimens prepared from samples taken at sixty-three sites in the Pre-Cambrian Giles Complex of Central Australia it was possible to recognize a stable magnetization. The results established that the Giles Complex was in its present position at the time of acquisition of its TRM. The magnetic properties tended to correlate with the state of exsolution and oxidation of the iron-titanium oxides in the rocks.

Introduction

The Pre-Cambrian Giles Complex is a series of layered mafic and ultramafic intrusions extending over an area of at least 12 000 sq km of Central Australia (Fig. 1). Nesbitt & Talbot (1966) have described the Complex. In addition to palaeomagnetic studies, it was expected that investigation of the magnetic properties of the rocks could provide information on the structural relationships (and history) of the Complex. Complementing magnetic studies, this investigation included study of certain petrological/mineralogical properties of these coarse-grained rocks.

Present weathering of the Giles Complex rocks is characteristic of arid regions, although chemical weathering has emphasized the strongly developed layering of the intrusions. The mafic portions tend only to have thin weathering rinds (at times less than a millimetre in thickness).

Magnetic investigation

Information gained during a preliminary study (Facer 1967) enabled a programme of sampling to be formulated. Where possible, sampling traverses were planned to be perpendicular to the strike of the layering of the Complex. Because natural surface water was virtually absent, and it was not practicable to have assistance in traversing the extremely rugged terrain, field coring procedures could not be employed in sampling. In addition to the sample orientation, at each site the orientation of the igneous layering was recorded, and at some sites dolerite dykes were also sampled. At least one thin-section and one polished surface were prepared from each sample.

As expected, directions of NRM showed considerable scatter (Facer 1970a). Preliminary Fisher (1953) analysis of NRM directions suggested that unit weight should be assigned to specimens rather than to samples. Since the NRM results appear to be of limited value for palaeomagnetic analysis, a number of ' stability tests ' were carried out.

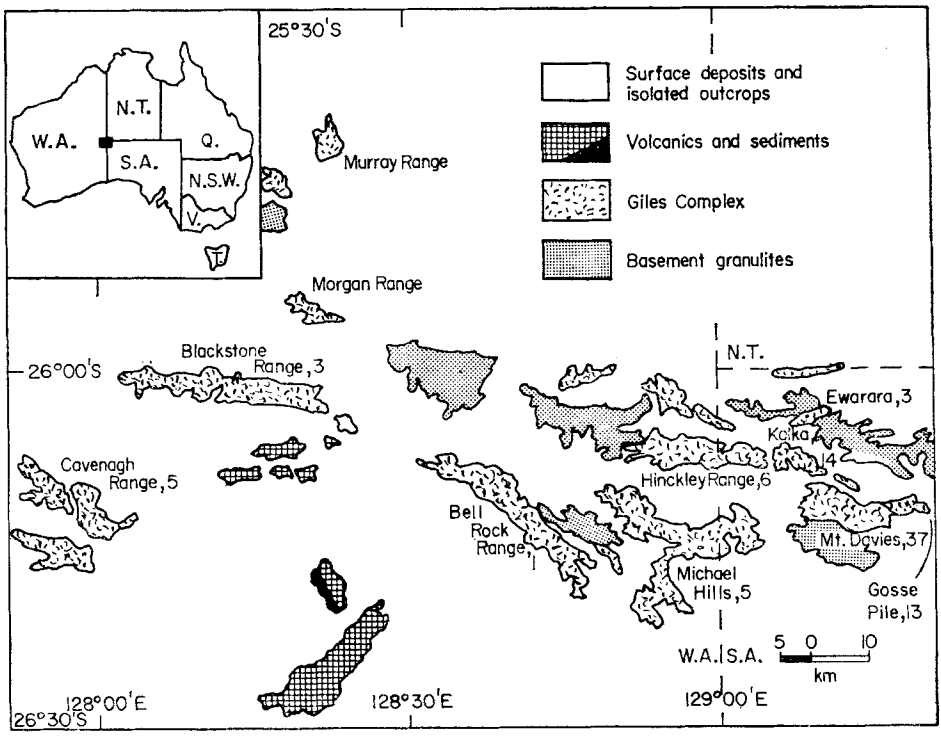


FIG. 1. Simplified geological map of the nine bodies of the Giles Complex sampled, based on Fig. 1 of Nesbitt & Talbot (1966) and on field observations. The number following the name indicates the number of oriented samples taken. The outcrops of the Giles Complex extend both to the East and West of the area shown here.

Shelf-tests over periods of up to two years showed that some samples, especially those of lower intensity, acquired a small VRM component in the laboratory. Both the Koenigsberger ratio and ' proportional Q-value ' of Zijderveld (1968, p. 3775) showed strong correlation with the partial AF demagnetization results. Additional procedures and detailed results will be given in a future paper by this author. The stability tests, together with very low intensities of magnetization of the ultramafic rocks, caused 53 per cent of samples to be rejected.

Very extensive Fisher (1953) analyses were carried out on various groupings of the stable directions after partial AF demagnetization using one of the computer program specifically written for this investigation. Unit weight was assigned to both samples and specimens. Table 1 summarizes these results, although those for the dolerite dykes have not been included. Rotation of the igneous layering into a horizontal orientation is an example of the use of Graham's fold-test, its application in this case having the effect of correcting for both tilting and broad warping. Using these results, the position of the North palaeo-pole corresponding to the Giles Complex (age apparently 1100 My) is Latitude 68° N, Longitude 17° W (semi-axes of the ellipse of 95 per cent confidence 23°, 29°) (Facer 1970a).

Study of the distribution of the directions of magnetization summarized in Table 1 showed that there was an even division of directions into normal and reversed groups, the mean direction for each group having overlapping circles of 95 per cent confidence (although of opposite polarity). Hence all reversed directions were inverted to normal and Fisher (1953) analyses repeated. Table 2 shows that there is noticeably tighter grouping about the mean direction of magnetization than was

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obtained for non-inverted directions (Table 1). The palaeo-pole position corresponding to the specimen mean direction before rotation was Latitude 34° N, Longitude 49° W (semi-axes of the near-circular ellipse of 95 per cent confidence 24°, 24°). Because the directions were evenly divided between normal and reversed, it was not possible to decide whether the palaeo-pole represented a normal or a reversed dipole field.

The direction means in Tables 1 and 2, apart from the sample units ' after rotation ' in Table 1, have values of R which indicate that ' the null hypothesis of randomness ' (Watson 1956, p. 161) can be rejected at both the 95 per cent *and* the 99 per cent levels for these means. It can be seen that specimen-unit weight means show tighter grouping than sample-unit weight means, an observation contrary to that of Roy (1966).

The petrological/mineralogical studies did not show any evidence of chemical alteration of opaque iron–titanium grains following high-temperature oxidation. Also a tendency for correlation between magnetic properties and mineralogy/petrology warrants further study in other coarse-grained mafic and ultramafic intrusive rocks (Facer 1970b).

Table 1

Fisher analysis of the Giles Complex Rocks which apparently possess stable TRM's

	Unit	No. of units	Declination of R	Inclination of R	R	α_{95}	k
Before	Sample	39	346 °T	+61°	15·6	29°	1·63
After	Sample	39	007 °T	+15°	11·8	36°	1·39
Before	Specimen	68	343 °T	+62°	30·8	19°	1·80
After	Specimen	68	005 °T	+21°	20·6	26°	1·41

Before: TRM directions before rotation of the layering to horizontal.
After: TRM directions with the layering horizontal.
R is the resultant vector, its direction representing the mean direction of the group.
α_{95} is the half-angle of the cone of 95 per cent confidence.
k is Fisher's estimate of precision.

Table 2

Fisher analysis of the Giles Complex Rocks which apparently possess stable TRM's—after inverting reversed directions to normal

	Unit	No. of units	Declination of R	Inclination of R	R	α_{95}	k
Before	Sample	39	145 °T	−86°	26·1	16°	2·95
After	Sample	39	335 °T	−26°	16·9	27°	1·72
Before	Specimen	68	169 °T	−86°	47·0	12°	3·19
After	Specimen	68	322 °T	−35°	27·6	21°	1·66

Before: TRM directions before rotation of the layering to horizontal.
After: TRM directions with the layering horizontal.
R is the resultant vector, its direction representing the mean direction of the group.
α_{95} is the half-angle of the cone of 95 per cent confidence.
k is Fisher's estimate of precision.

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Discussion

The apparent failure of Graham's fold-test (as shown in Tables 1 and 2) indicates that the nine Giles Complex bodies sampled were probably in, or very near, their present orientation at the time of formation of their stable magnetization. If, as seems likely, this is a TRM, the magnetic evidence shows that the Giles Complex was tilted before the rocks passed through their Curie temperature for the last time. Study of chemical analyses and polished surfaces indicated a low titanium content in the rocks, thus suggesting a moderate to high Curie temperature. Since the sampled rocks were probably intruded at depth (Goode & Krieg 1967; Moore 1968), and the gravitational layering was approximately horizontal, it is likely that the Giles Complex was tilted and uplifted soon after intrusion and consolidation, but *before* final cooling of the bodies. The present curvature of igneous layering, whether due to original orientation variation in the magma chamber or to folding, does not appear to have significantly influenced these conclusions.

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References

Facer, R. A., 1967. A preliminary study of the magnetic properties of rocks from the Giles Complex, Central Australia, *Aust. J. Sci.*, **30,** 237–238.

Facer, R. A., 1970a. Magnetic properties of the Giles Complex, Central Australia, *Search*, **1,** 76–77.

Facer, R. A., 1970b. Continental drift: Implications of paleomagnetic studies, meteorology, physical oceanography, and climatology: A discussion, *J. Geol.*, **78,** 630–633.

Fisher, R., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond*, **217A,** 295–305.

Goode, A. D. T. & Krieg, G. W., 1967. The geology of Ewarara intrusion, Giles Complex, Central Australia, *J. geol. Soc. Aust.*, **14,** 185–194.

Moore, A., 1968. Rutile exsolution in orthopyroxene, *Contr. Miner. Petrology*, **17,** 233–236.

Nesbitt, R. W. & Talbot, J. L., 1966. The layered basic and ultrabasic intrusives of the Giles Complex, Central Australia, *Contr. Miner. Petrology*, **13,** 1–11.

Roy, J. L., 1966. Désaimantation thermique et analyse statistique des directions de sédiments Carbonifères et Permiens de l'est du Canada, *Can. J. E. Sci.*, **3,** 139–161.

Watson, G. S., 1956. A test for randomness of directions, *Mon. Not. R. astr. Soc. geophys. Suppl.*, **7,** 160–161.

Zijderveld, J. D. A., 1968. Natural remanent magnetizations of some intrusive rocks from the Sør Rondane Mountains, Queen Maud Land, Antarctica, *J. geophys. Res.*, **73,** 3773–3785.

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